

Amir Mirzai Golpayegani

📍 Calgary, AB, Canada

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Summary

M.Sc. Computer Science graduate with experience building reliable Python/C++ software, REST APIs, data pipelines, and automation tools for complex technical workflows. Skilled in backend development, Git-based engineering, databases, documentation, testing-minded development, and learning new cloud technologies quickly in collaborative project environments.

Education

M.Sc. in Computer Science University of Calgary	GPA: 3.94/4.0	Sep 2022 – Feb 2026 Calgary, Canada
B.Sc. in Computer Engineering Sharif University of Technology	GPA: 3.69/4.0	Sep 2016 – Feb 2021 Tehran, Iran

Technical Skills

Programming: Python, C++, JavaScript, SQL

ML / AI: PyTorch, Scikit-learn, TensorFlow, CNNs, LLM workflows, AI agents, model evaluation

Data / Geospatial: NumPy, Pandas, SciPy, Google Earth Engine API, Discrete Global Grid System (DGGS), GDAL, Rasterio, SNAP

Backend / Engineering: Django, Node.js, REST APIs, PostgreSQL, Docker, Git, Linux, AWS, GitHub Actions, technical documentation

Visualization / Automation: Streamlit, Polyscope, OpenGL, GLSL, n8n, OpenClaw, SearXNG, Telegram API

Work Experience

Research Software Developer Sep 2022 – Feb 2026
BigGeo / Mitacs, University of Calgary Calgary, Canada

- Built C++/Python software for scalable processing, storage, and retrieval of Earth observation data in a collaborative research-industry project.
- Automated SAR and optical imagery preprocessing using GDAL, Rasterio, SNAP, and satellite data APIs.
- Developed DGGS-based encoding pipelines to convert satellite imagery into structured tensor files.
- Improved encoding performance with C++ multithreading, reducing 10-tensor processing from 233s to 26s.
- Documented data-processing methods, performance results, and implementation details for thesis and publication workflows.

Back-End Developer Jun 2020 – Jul 2022
Asr Gooyesh Pardaz Tehran, Iran

- Developed Django REST APIs for NLP, text-to-speech, and speech-to-text service plans backed by PostgreSQL.
- Built authentication features including registration, Google login, password reset, and email/phone verification.
- Implemented backend logic for payment-service integration and storing plan/payment records in a relational database.
- Collaborated on service features from requirements through implementation, debugging, and production-oriented support.

Deep Learning Teaching Assistant Jul 2026
Neuromatch Academy Remote, United States

- Mentored an international pod of ~15 students through Neuromatch Academy's intensive three-week Deep Learning course.
- Facilitated hands-on PyTorch tutorials spanning MLPs, CNNs, attention/transformers, diffusion generative models, and reinforcement learning.
- Guided project groups through end-to-end deep learning research projects, from question formulation and literature review to model implementation and final presentations.

Teaching Assistant Dec 2022 – Apr 2025
Department of Computer Science, University of Calgary Calgary, Canada

- **Developing Big Data Applications:** Mentored +25 students on AWS-based data pipelines, cloud storage, debugging, and scalable data workflows.
- **Working with Data at Scale:** Supported +20 students with large-scale data processing, data pipelines, and high-volume data analysis.
- **Database Management Systems:** +75 Assisted students with SQL, database design, course projects, grading, and exam support.

Research and Technical Projects

Spatiotemporal Earth Observation Data System Using DGGS Sep 2022 – Feb 2026
First-author publication | M.Sc. thesis | C++, Python, DGGS, Wavelets, B-splines, Time-Series

- Designed a DGGs-based framework for real-time multiresolution management of satellite imagery. - Developed a tensor-based storage model for direct access to arbitrary regions at multiple resolutions. - Implemented a triangular wavelet scheme to support multiresolution retrieval without image-pyramid redundancy. - Built and validated reusable software components for data storage, retrieval, and performance evaluation. - Retrieved 105,489 DGGs cells per time-series query in 13.9s, compared with about 220s using global retrieval.	
DEM Super-Resolution Framework <i>Python, PyTorch, CNNs, Scientific Imaging, Google Earth Engine API, Polyscope, Data Processing and Visualization</i>	May 2024 – Feb 2026
- Trained deep ResNet models for 4× DEM super-resolution, reconstructing 30m elevation data from 120m inputs. - Built a Streamlit pipeline using Rasterio to parallelize GEE DEM tile downloads and systematically collect and preprocess mountainous terrain data worldwide. - Implemented PyTorch training and evaluation workflows with terrain-aware metrics such as RMSE, slope, TRI, TPI, and wavelet-based errors. - Developed a Polyscope-based 3D visualization tool to render DEM tiles from multiple angles and compare reconstruction quality in 3D.	
Agentic Job Search Automation Platform <i>OpenClaw, n8n, LLMs, AI Agents, Docker, Workflow Automation, SearXNG, Telegram API</i>	May 2026 – Present
- Building an agentic AI system to search jobs, analyze job fit, and generate tailored application materials. - Designed LLM-based workflows for resume/job matching, skill-gap analysis, and application decision support. - Integrated Google Sheets, Google Drive, and Telegram API to track jobs and send automated application alerts. - Prototyped workflow automation in n8n and Docker before migrating toward a more flexible OpenClaw-based agent system.	
Persian Text Normalizer for NLP <i>Python, NLP, Text Processing</i>	Oct 2016 – Feb 2021
- Developed a Python-based text normalization toolkit to standardize Persian text for downstream NLP pipelines. - Implemented preprocessing rules for cleaning, formatting, and preparing Persian text data for language-processing tasks.	
Publication	
Amir Mirzai Golpayegani , Mahmudul Hasan, Faramarz F. Samavati. <i>Real-Time Multiresolution Management of Spatiotemporal Earth Observation Data Using DGGs</i> . Remote Sensing, 2025. doi:10.3390/rs17040570	
Certificates	
Supervised Machine Learning: Regression and Classification , DeepLearning.AI and Stanford University, Coursera	Oct 2024
Applied Machine Learning in Python , University of Michigan, Coursera	Jul 2020
Languages	
English: Full Professional Proficiency (TOEFL 115/120) Persian: Native Proficiency	
Achievement	
Teaching Assistant Excellence Laureate Department of Computer Science, University of Calgary — awarded for outstanding teaching support (link)	2024/2025
Ranked 17th Nationwide in Iran University Entrance Exam (Konkour) Top 0.001% among all participants	May 2016